

CHARACTERIZATION OF CONTINUOUS DISTRIBUTIONS BY PROPERTY OF CONDITIONAL EXPECTATION

Ismihan G. Bairamov & Ayşen Apaydın

Ankara University, Faculty of Science, Department of Statistics, 06100,
Tandogan, Ankara, Turkey

Key words: Characterization of distributions; conditional expectation; hazard rate function.

Summary: Some new characterizations of continuous distributions expressed in terms of conditional expectations are provided. Characterizations of continuous distributions in terms of regressions of order statistics are given.

1. Introduction

Let X be a random variable (r.v.) with absolutely continuous distribution function (d.f.) and probability density function (p.d.f.) f . Denote $\bar{F}(x) = 1 - F(x)$ and $g(x) = \frac{f(x)}{\bar{F}(x)}$, $\bar{F}(x) > 0$. If X is a nonnegative r.v., then $\bar{F}$ and g can be interpreted as survival and hazard (or failure) rate functions of X , respectively. Certain useful identities are well known, for example

$$\bar{F}(x) = \exp \left\{ - \int_0^x g(t) dt \right\} \quad (1.1)$$

(c.f., Barlow and Proschan, 1975, p. 53).

The survival and hazard rate functions play an important role in statistical theory of reliability. Another important function is the mean residual life

(MRL) function of X ,

$$\psi_F(x) = E(X - x | X \geq x) = \frac{1}{\bar{F}(x)} \int_x^\infty t f(t) dt - x, \quad \bar{F}(x) > 0,$$

which can be interpreted as the expected remaining life given survival at age x of an element having life length X with d.f. F and p.d.f. f . For details, c.f., e.g., Hall and Wellner (1981). The MRL function is related to other well-known functions such as the Lorenz curve and the hazard function (c.f., e.g., Arnold, 1983).

The hazard (or failure) rate function and mean residual life function are reliability concepts useful for the characterization of distributions.

Characterization of probability distributions has been investigated in numerous papers and monographs including Kagan, Linnik and Rao (1973), Galambos and Kotz (1978), Azlarov and Volodin (1986) and Arnold and Huang (1995).

Let $X_1, X_2, \dots, X_n$ be independent and identically distributed (i.i.d.) random variables with a common distribution function F . Denote by $X_{1:n} \leq X_{2:n} \leq \dots \leq X_{n:n}$ the order statistics constructed from $X_1, X_2, \dots, X_n$. The best unbiased predictor for $X_{m+k:n}$ given $X_{k:n}$ with respect to the squared-error loss is $E(X_{m+k:n} | X_{k:n})$. Historically Ferguson (1967) seems to be the first author to consider the problem of determining all d.f.s. F such that

$$E(X_{k+m:n} | X_{k:n}) = cX_{k:n} + d \text{ a.s.}$$

He provides a complete solution for $m = 1$. Wesolowski and Ahsanullah (1997) cite various authors who deal with similar problems. In the paper by Wesolowski and Ahsanullah, a solution for the case of absolutely continuous distributions with $m = 2$ is given. Blaquez and Rebollo (1997) obtain a

solution for the regression of $X_{k+m:n}$ on $X_{k:n}$, $1 \leq k < k+m \leq n$ for k times differentiable F .

Characterization of distributions in terms of conditional expectation and hazard rate function is given in Koicheva (1993), Ghitany, El-Saidi and Khalil (1995). Balasubramanian and Dey (1997) present results based on conditional expectation that lead to characterizations of several distributions. In their characterization they use harmonic and geometric means. One of the interesting contributions of this paper is the observation that characterization of distributions by regression of order statistics is closely related to characterization by properties of conditional expectation of some function of X under condition $x < X < y$, where x and y belong to the support of X .

Ghitany, El-Saidi and Khalil (1995) consider the general class of absolutely continuous distributions having p.d.f. of the form

$$f(x; \theta) = \exp[-q(x; \theta)], \quad x \geq 0, \quad \theta \in D \quad (1.2)$$

where $q(x; \theta)$ is a real-valued function with $q'(x; \theta) \neq 0$ and $q''(x; \theta)$ existing in $(0, \infty)$ for all $\theta \in D$. They prove the following theorem.

Theorem 0. Let X be a non-negative, absolutely continuous random variable with failure rate function $r(x; \theta)$. Then X has a p.d.f. given by (1.2) if and only if for a real-valued function $g(x; \theta) \neq 0$ possessing derivative on $(0, \infty)$ for all $\theta \in D$, the representation

$$E \left\{ \left[1 + \frac{q''(X; \theta)}{[q'(X; \theta)]^2} g(X; \theta) - \frac{g'(X; \theta)}{q'(X; \theta)} \right] | X \geq y \right\} = \frac{g(y; \theta)}{q'(y; \theta)} r(y; \theta) \quad (1.3)$$

holds for all $y \geq 0$.

Careful study of the proof of Theorem 0 reveals that the condition requiring $r(x; \theta)$ to be a failure rate function of X is essential; without it the proof is not valid. Consequently, when giving a characterization for some r.v. X expressed by conditional expectation using this Theorem we have to provide a condition that $r(y; \theta)$ appearing on the right-hand side of (1.3) is the failure rate function of X . But if the failure rate of X is known, then one can determine F by the formula (1.1). Hence, in such a characterization, (1.3) does not give any useful information, except a relationship between conditional expectation and a given hazard rate function. For the example given by authors, a characterization of the exponential distribution may be expressed as follows: Let $r(y; \beta)$ be the failure rate of X (consequently if, $r(y; \beta) = \beta$ then X is exponential). Then X has exponential distribution if and only if for an integer $k \geq 1$, the representation $E \left[X^{k-1} \left(X - \frac{k}{\beta} \right) | X \geq y \right] = \frac{y^k}{\beta} r(y; \beta)$ holds for all $y \geq 0$.

In this note we try to establish a theorem characterizing F in terms of conditional expectation without the restriction mentioned above. Using this result we give a characterization for continuous distributions in terms of the properties of regressions of order statistics.

2. The main result

Let X be a continuous random variable with the distribution function F . Denote by $F^{-1}(y) = \inf \{x : F(x) \geq y\}$, $0 < y < 1$ the inverse distribution function. Let $[a_X, b_X]$ ($-\infty \leq a_X < b_X \leq \infty$) be the endpoints of support of X , i.e. $a_X = \inf \{x : F(x) > 0\}$, $b_X = \sup \{x : F(x) < 1\}$. It is well known that the random variable $F(X)$ has a uniform distribution on $(0, 1)$. It is also true that if U is a random variable with a uniform distribution on

$(0, 1)$ then the r.v. $X = F^{-1}(U)$ has d.f. $F(x)$ (see, e.g. Randles and Wolfe, 1979).

Theorem 1. Let X be a continuous r.v., $f(x)$ be a differentiable real valued function, such that $h(x) < \infty$ and

$$h'(x) \neq \frac{h(1) - h(x)}{1 - x} \text{ for all } x \in (0, 1). \quad (2.1)$$

Then X has the distribution function $G(x)$ if and only if the representation

$$E \{h'(G(X)) | G(X) \geq z\} = \frac{h(1) - h(z)}{1 - z} \quad (2.2)$$

holds for all $z \in (0, 1)$.

Proof. It is clear that (2.2) holds for all $z \in (0, 1)$ if and only if

$$E \{h'(G(X)) | X \geq y\} = \frac{h(1) - h(G(y))}{1 - G(y)} \quad (2.3)$$

holds for all $y \in (a_X, b_X)$.

Necessity: Let X have d.f. G . Then, letting $G'(x) = g(x)$ one can write

$$\begin{aligned} E \{h'(G(X)) | X \geq y\} &= \frac{1}{\bar{G}(y)} \int_y^{b_X} h'(G(x)) g(x) dx \\ &= \frac{1}{\bar{G}(y)} \int_y^{b_X} dh(G(x)) \\ &= \frac{1}{\bar{G}(y)} [h(G(b_X)) - h(G(y))] \\ &= \frac{h(1) - h(G(y))}{\bar{G}(y)}. \end{aligned}$$

Sufficiency: Let (2.3) hold. Denote the d.f. and p.d.f. of X by F and f ,

respectively. Then we have from (2.3)

$$\frac{1}{\bar{F}(y)} \int_y^{b_X} h'(G(x)) f(x) dx = \frac{h(1) - h(G(y))}{\bar{G}(y)} \quad (2.4)$$

Rewrite (2.4) as

$$\int_y^{b_X} h'(G(x)) f(x) dx = \frac{\bar{F}(y)}{\bar{G}(y)} [h(1) - h(G(y))] \quad (2.5)$$

Differentiating (2.5) with respect to y in $(0, b_X)$ one obtains

$$\begin{aligned} -h'(G(y)) f(y) &= \frac{-h'(G(y)) g(y) \bar{G}(y) + g(y) (h(1) - h(G(y)))}{(\bar{G}(y))^2} \bar{F}(y) \\ &\quad - f(y) \frac{h(1) - h(G(y))}{\bar{G}(y)}. \end{aligned} \quad (2.6)$$

It follows from (2.6) that

$$\begin{aligned} &h'(G(y)) \frac{1}{\bar{F}(y) \bar{G}(y)} (g(y) \bar{F}(y) - f(y) \bar{G}(y)) \\ &= (h(1) - h(G(y))) \frac{1}{\bar{F}(y) \bar{G}^2(y)} (g(y) \bar{F}(y) - f(y) \bar{G}(y)) \end{aligned} \quad (2.7)$$

From (2.7) one obtains

$$(g(y) \bar{F}(y) - f(y) \bar{G}(y)) \left(h'(G(y)) - \frac{h(1) - h(G(y))}{\bar{G}(y)} \right) = 0. \quad (2.8)$$

Consider (2.8). If $g(y) \bar{F}(y) - f(y) \bar{G}(y) = 0$ for all y , then

$$\frac{g(y)}{\bar{G}(y)} = \frac{f(y)}{\bar{F}(y)}, \quad \text{i.e. } F = G.$$

Suppose now that there exists $y_0 \in (a_X, b_X)$, such that

$$g(y_0) \bar{F}(y_0) - f(y_0) \bar{G}(y_0) \neq 0$$

Then from (2.8) one can write

$$h'(G(y_0)) - \frac{h(1) - h(G(y_0))}{\bar{G}(y_0)} = 0, \quad (2.9)$$

Taking $G(y_0) = x_0$ in (2.9) one has

$$h'(x_0) - \frac{h(1) - h_1(x_0)}{1 - x_0} = 0, \quad (0 < x_0 < 1)$$

which contradicts the condition of the theorem. The theorem is thus proved.

Let $X_1, X_2, \dots, X_n$ be a random sample from the distribution $G(\cdot)$ of the variable X defined in Theorem 1 and $X_{1:n}, X_{2:n}, \dots, X_{n:n}$ be the corresponding order statistics. It can easily be verified that the conditional distribution of $X_{r+1:n}, X_{r+2:n}, \dots, X_{n:n}$ given $X_{r:n} = x$ is the same as the distribution of $Y_{1:n-r}, Y_{2:n-r}, \dots, Y_{n-r:n-r}$, where $Y_{i:n-r}$ is the i th order statistic of the sample of size $n - r$ obtained from the truncated distribution $\{X \mid X \geq x\}$. Using Theorem 1 we have the following:

Theorem 2. Under the condition of Theorem 1, X has distribution function $G(x)$ if and only if the representation

$$E \left\{ \frac{1}{n-r} \sum_{i=r+1}^n g(X_{i:n}) \mid X_{r:n} = y \right\} = \frac{h(1) - h(G(y))}{1 - G(y)}$$

holds for all $a_X < y < b_X$, where $g(x) \equiv h'(G(x))$.

3. On selection of $h(x)$

1. Let $n \geq 1$ and $h(x) = \frac{x^{n+1}}{n+1}$. Then $h'(x) = x^n$. Consider the equation

$$\frac{h(1) - h(x)}{1 - x} = h'(x), \text{ i.e. } \frac{1 - x^{n+1}}{n+1} = (1 - x)x^n \quad 0 < x < 1. \quad (3.1)$$

One can write from (3.1)

$$\begin{aligned} 1 - x^{n+1} &= (n+1)x^n - (n+1)x^{n+1} \Rightarrow (n+1)x^n - nx^{n+1} - 1 = 0 \\ &\Rightarrow nx^{n+1} - (n+1)x^n + 1 = 0. \end{aligned} \quad (3.2)$$

It is clear that for a given finite n equation (3.2) has no solutions in $(0, 1)$.

Thus $h(x)$ satisfies the condition of Theorem 1.

2. Let

$$\begin{aligned} h(x) = & -(1-x) \left\{ (\ln(1-x))^n - n(\ln(1-x))^{n-1} \right. \\ & + n(n-1)(\ln(1-x))^{n-2} - n(n-1)(n-2)(\ln(1-x))^{n-3} \\ & \left. + \dots + (-1)^n n! \right\} \end{aligned} \quad (3.3)$$

One can check that $h'(x) = (\ln(1-x))^n$. Denoting $y = \ln(1-x)$ in the equation $\frac{h(1)-h(x)}{1-x} = h'(x)$ one has $y^n - ny^{n-1} + n(n-1)y^{n-2} - n(n-1)(n-2)y^{n-3} + \dots + (-1)^n n! = y^n$ which can be rewritten as follows

$$ny^{n-1} - n(n-1)y^{n-2} + n(n-1)(n-2)y^{n-3} + \dots + (-1)^n n! = 0.$$

The last equation has no solutions in $(0, 1)$.

4. Characterizations for some continuous distributions

In this section we give some examples to illustrate the use of Theorems 1 and 2 for characterizations of distributions.

1. *Uniform $[0, 1]$ distribution.*

Let

$$F(x), \quad 0 < x < 1.$$

Taking

$$h(x) = \frac{x^{n+1}}{n+1}, \quad h'(x) = x^n$$

one obtains

$$\frac{h(1) - h(x)}{1 - x} = \frac{\left(\frac{1}{n+1} - \frac{x^{n+1}}{n+1} \right)}{1 - x} = \frac{1 - x^{n+1}}{(n+1)(1-x)}. \quad (4.1)$$

It is clear that (4.1) can be written as follows:

$$\frac{h(1) - h(x)}{1 - x} = \frac{1}{n+1} (1 + x + \dots + x^n) \quad (4.2)$$

Using Theorem 1 and taking into consideration (4.1) or (4.2) we have the following:

Corollary 1. The continuous random variable X has a *Uniform* $[0, 1]$ distribution if and only if the representation

$$E \{ X^n \mid X \geq x \} = \frac{1}{n+1} (1 + x + \dots + x^n)$$

holds for all $0 < x < 1$ and some $n \geq 1$.

Corollary 1A. The continuous random variable X has *Uniform* $[0, 1]$ distribution if and only if

$$E \left\{ \frac{1}{n-r} \sum_{i=r+1}^n X_{i:n}^k \mid X_{r:n} \right\} = \frac{1}{k+1} (1 + X_{r:n} + \dots + X_{r:n}^k)$$

for all $0 < x < 1$ and some $k \geq 1$, where $X_{i:n}$ is the i th order statistic of the sample of size n from X .

2. Weibull distribution.

Let

$$F(x) = 1 - e^{-\beta x^\alpha}, \quad x \geq 0, \quad \beta > 0, \quad \alpha > 0. \quad (4.3)$$

Taking $h(x)$ as in (3.3) one has $h'(x) = (\ln(1-x))^n$ we have the

following characterization of the Weibull distribution.

Corollary 2. The continuous and nonnegative random variable X has a Weibull distribution with d.f. (4.3) if and only if the representation

$$\begin{aligned} E\{X^{\alpha n} \mid X \geq y\} &= y^{n\alpha} + \frac{n}{\beta} y^{(n-1)\alpha} + \frac{n(n-1)}{\beta^2} y^{(n-2)\alpha} \\ &\quad + \frac{n(n-1)(n-2)}{\beta^3} y^{(n-3)\alpha} + \dots + \frac{n!}{\beta^n} \end{aligned}$$

holds for some $n \geq 1$ and for all $0 < y < \infty$.

Corollary 2A. The continuous and nonnegative random variable X has a Weibull distribution with d.f. (4.3) if and only if

$$\begin{aligned} E\left\{\frac{1}{n-r} \sum_{i=r+1}^n X_{i:n}^{\alpha m} \mid X_{r:n}\right\} &= (X_{r:n})^{m\alpha} + \frac{m}{\beta} (X_{r:n})^{(m-1)\alpha} \\ &\quad + \frac{m(m-1)}{\beta^2} (X_{r:n})^{(m-2)\alpha} + \dots + \frac{m!}{\beta^m} \end{aligned}$$

for some $m \geq 1$, where $X_{i:n}$ is the i th order statistic of the sample of size n from X .

3. Pareto distribution

Consider the Pareto $(\theta; 0, 1)$ distribution function $F(x, \theta) = 1 - x^{-\theta}$, $\theta > 0$, $x > 1$.

Let us take

$$h(x) = -\frac{(1-x)^{n+1}}{n+1}, \quad h'(x) = (1-x)^n.$$

Then we have the following characterization for the Pareto distribution.

Corollary 3. The continuous r.v. X has a Pareto $(\theta; 0, 1)$ distribution

iff for some $\theta > 0$ the representation

$$E \{ X^{-n\theta} \mid X \geq y \} = \frac{y^{-n\theta}}{n+1}$$

holds for some $n \geq 1$ and for all $1 < y < \infty$.

Corollary 3A. The continuous r.v. X has a Pareto $(\theta; 0, 1)$ distribution if and only if

$$E \left\{ \frac{1}{n-r} \sum_{i=r+1}^n X_{i:n}^{-m\theta} \mid X_{r:n} \right\} = \frac{X_{r:n}^{-m\theta}}{m+1}$$

for some $m \geq 1$, where $X_{i:n}$ is the i th order statistic of the sample of size n from X .

It is evident that using the main theorem established in this work some other distributions of interest can also be characterized in a similar way by a judicious choice of the “target function” $h(\cdot)$.

Acknowledgement

The authors thank a referee for valuable suggestions which resulted in the improvement of the presentation of this paper. We appreciate the valuable comments of Professor Samuel Kotz.

References

- ARNOLD, B.C. (1983). *Pareto distributions*. Co-operative Publishing House, Fairland, Maryland.
- ARNOLD, B.C. & HUANG, J.S. (1995). Characterizations. *The Exponential Distribution. Theory, Methods and Applications*. Edited by N. Balakrishnan & Asit P. Basu. Gordon and Breach Publishers. 185–203.
- AZLAROV, T. & VOLODIN, N. (1986). *Characterization problems associated with the exponential distribution*. Springer-Verlag. Berlin.

- BALASUBRAMANIAN, K. & DEY, A. (1997). Distributions characterized through conditional expectations. *Metrika*, **45**, (3), 189–196.
- BARLOW, R.E. & PROSCHAN, F. (1975). *Statistical theory of reliability and life testing*. Holt, Rinehart & Winston, New York.
- BLAQUEZ, F.L. & REBOLLO, J.L.M. (1997). A characterization of distributions based on linear regression of order statistics and record values. *Sankhya*, Vol. 59, Series A, Pt. 3, pp. 311–323.
- FERGUSON, T.S. (1967). On characterizing distributions by properties of order statistics. *Sankhya*. Ser. A., **29**, 265–278.
- GALAMBOS, J. & KOTZ, S. (1978). *Characterizations of probability distributions*. Springer, Berlin, Heidelberg, New York.
- GHITANY, M., EL-SAIDI, A. & KHALIL, Z. (1995). Characterization of a general class of life-testing models. *J. Appl. Prob.*, **32**, 548–553.
- HALL, W.J. & WELLNER, J.A. (1981). Mean residual life. In *Statistics and Related Topics*. (Eds., M. Csörgö, D. Dawson, J.N.K. Rao & A.K. Md. E. Saleh), 169–184, North-Holland, Amsterdam.
- KAGAN, A.M., LINNIK, Yu. V. & RAO, C.R. (1973). *Characterization problems of mathematical statistics*. Wiley, New York.
- KOICHEVA, M. (1993). A characterization of the gamma distribution in terms of conditional moments. *Appl. Math.* **38**, 19–22.
- WESOŁOWSKI, J. & AHSANULLAH, M. (1997). On characterizing distributions via linearity of regression for order statistics. *Austral. J. Statist.* **39**(1), 69–78.